Humboldt University of Berlin

Long-run dynamics of wealth inequality
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Spatial Income Inequalities in Poland after 1990

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1. Introduction

Within one generation, Poland has moved from being one of the most egalitarian to one of the most unequal countries in Europe according to the results of Bukowski and Novokmet (2021). The top 1% income share in Poland during the 20th century has followed a U-shape similar to the one in the top developed economies, studied by Piketty and Zucman (2014). Before the Second World War, the top 1% income share in Poland was at a similar level as the one in the United Kingdom, France, Germany and Sweden, so between 8% (just after the First World War) and 20%. Under the communist regime from 1945 to 1989, Poland has seen its income inequalities fall. The top 1% income share varied between 5% and 4% between 1955 and 1989. After the abolition of communism, this share rose again, picking at 15% in 2007, and since then at the same level as in Germany and Sweden (around 13%). Similarly, the top 10% income share has risen continuously from 21% in the 1980's to 35% in the 2010's and the share of the bottom 50% income share has know a continuous drop from 32% to 24% over the same period (Bukowski et al. (2023)). Strikingly, the increase of the top income share was more severe in Poland as in other former satellite states of the USSR as Czechia and Hungary (Bukowski and Novokmet (2021)).

We propose to explore the role that regional disparities play in the return of income inequalities in Poland since the abolition of Communism. We find evidence in the Polish regional macroeconomic indicators that the income inequalities may be driven by regional inequalities. Indeed, the part of actual Poland that was Prussian from 1866 to 1914, which corresponds mostly to the West of the country (see appendix 11), presents overall better performance in terms of GDP per capita and risk of poverty[1]. However, the unemployment rate and expenditures in R&D in relation to GDP - a proxy of economic dynamism - are not as clearly distributed among the regions. Therefore, between region disparities don't seem to drive all of the return of income inequalities in Poland. In particular, what seems to drive the performance of the indicators cited above in the Mazowieckie Voivoidship is the city of Varsow (see appendix 12). This gives the insight that the Voivoidships - the most macro administrative regions of Poland, 16 in number since 1993 - may be a too broad scale to capture the main drivers of inequality. The income inequality within regions, between the locations driving economic growth and the ones less dynamic in this regard, is an aspect who deserves to be addressed by using a narrow observation scale and estimating the importance of the within observation units inequalities.

Income inequality in Poland is the subject of many articles by Filip Novokmet and Pawel Bukowski (Bukowski and Novokmet (2021), Bukowski et al. (2023), Bukowski and Novokmet (2024)). They are both professors and researchers in economics and political economics with a special focus on income inequalities in Eastern Europe countries. In their article of 2021, they present the result of their exploration of income group share based on long run data

[1] source : GUS, consulted on the 02/03/2026

never used before (aggregating Tax Statistics, Household Budget Survey, Personal Income Tax, EU Statistics on Income and Living conditions and National Accounts data). This is how they uncover the U-shape curve of the top percent income group over the 20th century. Focussing on the drivers of the return of income inequalities in Poland in their article published in 2023, Bokowski and Novokmet have shed light on the growth of business income at the top of the income distribution and the concurrently regressivity of the tax system at its very top. In their last work, making use of a RIF regression, they stress that in the decade following the fall of the wall, approximately 25% of the increase in income inequality can be ascribed to shifts in population composition (driven by an increase in the general level of education and the industrialisation of the economy), and the remaining 75% is due to changes in how these characteristics affect income distribution (the income structure effect). But these explanations don't address the regional-driven rise of income inequality since the 1990's. Only briefly addressed regarding the top 1% and top 0.1% in their 2023 article, the question of spatial wage inequalities appears to us as a key explanation factor of the U-shape curve in Poland.

Regional inequality within countries is a growing concern as economic geographic disparities within countries drive political tensions, increasing anti-system voting as an act of revenge against the city elites (Dijkstra et al. (2020)) or as a response to a feeling of abandonment (Piketty and Cagé (2023)). Research on regional inequality has gained traction and has demonstrated the rising disparity between sub-national regions. The strands of the literature on the subject are mostly grouped around three tendencies. The first one focusses on GDP per capita at a large sub-national level (NUTS1/TL1 or NUTS2/TL2 level) over long periods. They provide support for regional inequality N-shaped curve, arguing that spatial inequality follow an inverted U curve in the first stages of development and during periods of industrialization (recalling a Kuznet's curve), but with re-rising spatial inequality when the level of economic development is high (Lessmann (2014), Lessmann and Seidel (2017)). The second strand focuses on inequalities within local labour markets and cities. This is what has been achieved, adding a cross-country comparison aspect, by Bauluz et al. (2023) in the American, Canadian, British, French and German cases. The third strand of literature utilises administrative micro-level data to analyse the trends and sources of productivity and wages dispersion across areas. On of the main outcome of these studies is to show that high-skilled workers tend to concentrate in large, highly productive and high-wage locations (Card et al. (2025) in the case of the US; Roca and Puga (2017) in the case of Spain; Dauth et al. (2018) in the case of Germany). This mainly in this third vein that our work takes place, taking the net cash income for interest variable.

After presenting our data in section 2, and our methods in section 3, we present our findings in 4. We present first the descriptive statistics of income distribution in Poland over 1986-2023, and then the results of the Theil index at the Voivoidship and city size level.

2. Data

A primary contribution of our research is the novel utilization of survey data from the Public Opinion Research Center (CBOS). Although CBOS data have been previously employed in sociological studies to analyze attitudes toward inequality and subjective well-being (Grosfeld and Senik, 2010), they have not been systematically applied to the empirical estimation of regional income distributions. By comparing the CBOS data with traditional Household Budget Surveys and performing regional poststratification weighting, this study provides a more granular and frequent assessment of the spatial dynamics of inequality than has been previously possible in the Polish context.

2.1. CBOS Survey

Centrum Badania Opinii Społecznej (CBOS; eng.: *Centre for Public Opinion Research*) is Poland's leading public-opinion research institute, established in 1982. Since 1997, CBOS is sanctioned by law and is supervised a board of experts. It acts as a think tank to the Polish Parliament and Government (CBOS, 2026). In this paper we analyse monthly microdata of the CBOS's main statutory survey *Aktualne Problemy i Wydarzenia* (eng. *Current Problems and Events*) from January 1990 till December 2017[2]. The data has been downloaded from the Social Data Repository (RDS). Every survey was conducted on a representative sample of around 1000 people drawn from the central administrative PESEL system (eng. *Universal Electronic System for Registration of the Population*) and weighted. The post-stratification country-level weight has been computed using RIM Weighting with respect to gender, age group, type of locality and voivodeship, highest attained education and social or labour group (Fundacja Centrum Badania Opinii Społecznej, 2017).

The frequency of the data collection is a significant advantage of this dataset. It allows for a country-level analysis of the fluctuations of income and income inequalities within each year. Due to the monthly sample size it is not possible to draw any meaningful conclusions about spatial inequalities with each month. Therefore, we will pool the surveys into yearly batches for the regional analysis. A significant disadvantage of this dataset is the high missingness inside certain variables. We will investigate this issue closely in the section 3.3.

2.2. Household Budget Survey

The Household Budget Survey has been conducted by Statistics Poland (Główny Urzad Statystyczny, GUS) since 1986. It is conducted yearly since 2004, but it is only available for some year points before (1986, 1992, 1995, 1999). We have get access to these data

[2]In five months the survey has not been conducted. The open access repository is missing microdata for additional two months. In total, the dataset includes 329 surveys with 358,525 respondents.

through the Luxembourg Income Study Database (LIS). It is between 28 000 and 38 000 households that were interrogated each year. Descriptive statistics of the HBS data given the main outcome variables are available in appendix figure 2.

The main interest of this datasets is to put in perspective our results using the CBOS survey and also to have a glimpse into the shift in inequalities underwent in Poland as the communist regime collapsed. The survey has underwent important changes after 1990, becoming fully representative of the population according to Bukowski and Novokmet (2021). Therefore, we have paid particular attention to reweighing the variables as for the CBOS data, using the same RIM Weighting method. The HBS maintains a consistent two-stage stratified sampling scheme, with the main sample composed of two sub-samples, each of which is completely renewed every other year, while the other is re-used for the second time. Moreover, the methodology specifically mitigate the potential bias of under-representation of high-income households in self-reported surveys from 2020 onwards. From this year, the survey utilizes 26 'wealthier' strata—identified via tax records—which are oversampled by a factor of 1.2. The very rich are hence supposed to be better captured from 2020 onwards, which could lead to a slight overestimation of the increase of income inequalities in 2020, but we mitigate this by comparing the HBS data with the CBOS ones.

2.3. Income definition

The definition of income we focus on is largely dependent on the one of the income variables of our datasets. In the case of CBOS, the income amount value is asked at the personal and household level since 2000. It was before asked only at the household level. At the personal level, the question is asked to people aged 18 or older. One main issue when dealing with the CBOS data for income level analysis is that the question phrasing and its answering modalities are not exactly the same for each wave, which can slightly blur the results and the long term analysis of the data. Income value can be asked in threshold or in its precise amount. In February 1990, the question on household income is formulated as follows : "What is the NET monthly income per person in your household in NEW ZLOTYS? - average for the last 3 months"[3]. Later on, the question on income precise : "Please take into account your income from your main job, including bonuses, awards, allowances, income from additional work, including casual work, pensions and retirement benefits, scholarships, and other additional income, and calculate the average for the last 3 months"[4]. Therefore, we cannot reliably differentiate between sources or types of income and the phrasing supposes that the capital income are not taken into account. However,

[3]Originaly : Ile wynosza W NOWYCH ZŁOTYCH miesieczne dochody NETTO na 1 osobe w Pana(i) gospodarstwie domowym? - średnia za ostatnie 3 miesiace.

[4]Originaly : Prosze wziać pod uwage dochody z pracy głównej wraz z premiami, nagrodami, dodatkami, dochody z prac dodatkowych, także dorywczych, renty i emerytury, stypendia oraz inne odatkowe dochody wszystkich członków Pana(i) gospodarstwa domowego i obliczyć średnia za ostatnie 3 miesiace.

when the specific capital income question is mentioned (as in the February 1999 CBOS survey), only 5 respondents over more than 1000 reported non-null capital income. We face here the usual under-declaration of capital income, but crossing our results with CBOS with the ones of HBS, we try to bring a more robust analysis of income inequality.

Considering HBS, income at the household level takes into account the capital income, which the variable at the personal doesn't do (for survey construction reasons). Income at the individual level considers individuals over 15 years of age, age at which the individuals can legally begin to work for pay. Because the capital income are main drivers of income inequality, the results at the household level will take a more important place than the ones at the personal level. The non-capital income considered are cash incomes from labour (including wage income, self-employment income, but excluding own consumption and fringe benefits), from pensions (including both public and private pensions) and non-pension public social benefits whose eligibility is based on individual rather than household characteristics (namely maternity and parental leave, unemployment benefits, sickness and work injury pay and disability benefits). At the household level, some additional public social benefits are included (child allowance, general assistance, housing benefits and public in kind benefits).

In both cases, all incomes are net of taxes and debts. The flaws of each income variable are mitigated by our cross-dataset analysis but they are important to keep in mind to understand well the results.

3. Methods

3.1. Income shares

Following the tradition of the income inequality study papers (Alvaredo et al. (2018), Chancel et al. (2022)), our major way to analyse income is by sampling the population into income groups. Specifically, we take the quantiles of the income distribution in the reference population to distinguish the relevant social groups. We have chosen to constitute a "lower class" group with the bottom 50% of the income distribution, a "middle class" group with the middle 40% (ie. with the people whose income is between the 5th decile to the 9th decile), an "upper-class group" with the top 10% and a "very-upper class" group with the top 1% people of the income distribution. We then measure the mean income within each group and, in order to compute the income shares of this social groups, the ratio of these values on the mean income of the total population. We didn't get further into the decomposition of the top income fractions for data reasons. These people incomes are hard to catch with surveys as the ones we use and are mainly available in fiscal data, which are not in our disposition.

The benefit of this approach by income shares is to highlight the income gap between groups of a given population without concerns on the general change of income. It is so an advantageous analysis method for long-term study of income inequality. It is not about the total creation of wealth but on its distribution that this method focusses on (even if the two questions are related).

3.2. Regional weighting framework

In order to conduct a reliable regional income analysis, we have to adjust the survey weights to achieve regional and locality size representativeness. Initially, both surveys are weighted using GUS demographic data that is based on censuses. In this paper, we also employ GUS demographic data and build a quasi-continuous framework using both marginal and joint distributions of multiple demographic characteristics. We classify the demographic datasets into two groups:

- Structural seeds – census tables that provide highly reliable data on the most granular spatial level.
- Dynamic constraints – annual demographic data for territorial units higher order.

The main goal is to interpolate the seed tables under the time specific dynamic constraints, to achieve a detailed overview of annual demographic structure of Polish regions from 1986 to 2023. It is important to note, that all currently digitalized GUS data is harmonized for the new administrative division of Poland (16 voivodeships) implemented in 1999 (see appendix A). In order to aggregate the data into 49 pre-reform voivodeships, we produce a mapping between post-reform communes and the old territorial division. To this end, we overlay high resolution shapes of former voivodates on the shapes of communes from similar period of time. Exactly this process is the reason why the interpolation happens primarily on the communal level. Extraction of this mapping is crucial for the interpolation in the early period (1986-1994), since we can utilize historical data to establish hierarchical constraints.

The backbone of the algorithm is the census data (1988, 2002, 2011 and 2021) on the settlement aggregation level. Besides that, we use yearly (1995-2024) demographic data provided by the GUS' Local Data Bank on the commune and county aggregation levels. The dynamics of the education groups are modelled with respect to the voivodeship-level data of the BAEL study (eng. *Labour Force Survey*) conducted yearly by GUS starting from 1992. This data constraints the commune-level estimates between structural seeds. To establish hierarchical constraints and the dynamics in early phase (1986-1994) of the interpolation, we digitalize (old) voivodeship-level cross tables for age groups and gender, and marginal distributions of education groups from Statistical Yearbooks of GUS. The detailed overview of the auxiliary data is available in the appendix B.3.

The interpolation happens in three steps:

1) For every census anchor we compute log-linear interaction parameters for each commune's cross-tabulations. We interpolate the missing parameters in log-space piecewise with cubic (Hermite) splines and achieve seed tables carrying the association structures.

2) For each missing year with known margins we apply multi-dimensional Iterative Proportional Fitting to adjust the seed table to the marginal distributions. If the margins are unknown (only early period), we use the seed table directly.

3) We verify if the commune-level estimates aggregate consistently within their higher administrative divisions to the values of their hierarchical parents. If that's not the case, we adjust the values by jointly minimizing the deviation of the values from the values of the appropriate higher territorial unit.

A similar approaches has been successfully implemented for census data; see e.g. Beckman et al. (1996). After building the cross-tabulations both datasets has been reweighed using Iterative Proportional Fitting (raking) with the tables aggregated to the appropriate level: old or new voivodeship, macroregion, locality class.

3.3. Data handling and the issue of missing data

Both datasets has been harmonized to two unified labelling systems. This approach has been imposed by the construction of the weighting framework. Overview available in Appendix.

The CBOS dataset has been cleaned by excluding all respondents with missing data about age, gender, locality size, voivodeship and highest attained education, reducing the total number of observations from 358,525 to 355,352. These variables are populated in the most cases at the time of drawing the sample and are only verified with the respondents. The issue of missing data arises for the income variables with 55.3% of the observations missing for the personal income variable and 22.73% for the income per household member. When personal income is missing the household-member measure is observed in 73.87% of instances, and when the household-member measure is missing personal income is observed in 36.42% of instances. Both variables are missing for the 14.45% of all respondents. In 1.76% of the cases when household p.c. income is missing, the household consists only from one person and we can directly use the observed personal income.
In order to analyse the issue of missingness in detail, we follow a similar approach as described in Allison (2009) and fit per-survey logistic regressions predicting the missingness from covariates in two settings:

- Model A: age, gender, locality size, education and household size
- Model B: Model A and subjective standard of living.

The results of the analysis in appendix B.2 suggest that MCAR can be completely rejected.

Observed covariates meaningfully predict missingness (moderate values of AUC). Values of the effect sizes imply that younger respondents tend to disclose both types of income more often than older participants. Members of bigger households are less likely to report values of both variables. This is a plausible result, since the respondent might not know the incomes of all other household members. Finally, the data has been completed using Multivariate Imputation by Chained Equations based on the Model B (see Fig. 9).

3.4. Theil index

The Theil index is a statistic proposed by the Dutch econometrician Henri Theil. Its principle comes from information theory: it computes the maximum possible entropy of the data minus the observed entropy. Entropy quantifies the average level of uncertainty or information associated with the variable's potential states or outcomes. In our case, the maximum possible entropy would be a perfect equality of incomes among individuals and the observed entropy represents how randomly income is distributed. Therefore, when the Theil index approaches 0, we are close to a perfect egalitarian setting, and vice-versa when it approaches 1.

Denoting T_T the value of the Thiel index, N the number of individuals, R the number of regions, y_i the income of individual i, μ the mean income of total population and Y the total income earned by the population ($Y = N\mu$), we have :

$$T_T := \frac{1}{N}\sum_{r=1}^{R}\sum_{i\in r}\frac{y_{ir}}{\mu}\ln\left(\frac{y_{ir}}{\mu}\right) = \frac{1}{Y}\sum_{r=1}^{R}\sum_{i\in r} y_{ir}\ln\left(\frac{y_{ir}}{\mu}\right). \tag{1}$$

In other terms, it computes the average of the logarithms of the reciprocals of income shares weighted by income.

Using geolocated data, we can decompose the Theil index into two parts: a within component and a between component. This is of particular interest in our case, as we are interested in income inequality within and between Polish regions. Denoting N_r the number of individuals in region r, μ_r the mean income in region r, Y_r the total income earned by the population in region r and s_r the income share of region r ($s_r = \frac{N_r\mu_r}{N\mu}$), we have

$$T_T = T_{\text{between}} + T_{\text{within}} \tag{2}$$

so that,

$$T_{\text{between}} = \sum_{r=1}^{R}\frac{N_r\mu_r}{N\mu}\ln\left(\frac{\mu_r}{\mu}\right) = \sum_{r=1}^{R} s_r\ln\left(\frac{\mu_r}{\mu}\right), \tag{3}$$

and

$$T_{\text{within}} = \sum_{r=1}^{R} \frac{N_r \mu_r}{N\mu} \left[\frac{1}{Y_r} \sum_{i \in r} y_{ir} \ln\left(\frac{y_{ir}}{\mu_r}\right) \right]. \tag{4}$$

With $\frac{1}{Y_r} \sum_{i \in r} y_{ir} \ln\left(\frac{y_{ir}}{\mu_r}\right) = T_{T_r}$, the Theil index for the region r subgroup, we have

$$T_{\text{within}} = \sum_{r=1}^{R} s_r T_{T_r}, . \tag{5}$$

Finally, injecting (3) and (5) into (2), we have

$$T_T = \sum_{r=1}^{R} s_r \ln\left(\frac{\mu_r}{\mu}\right) + \sum_{r=1}^{R} s_r T_{T_r}. \tag{6}$$

Although the Theil index is always between 0 and 1, the contributions to the Theil Index may be negative or positive.

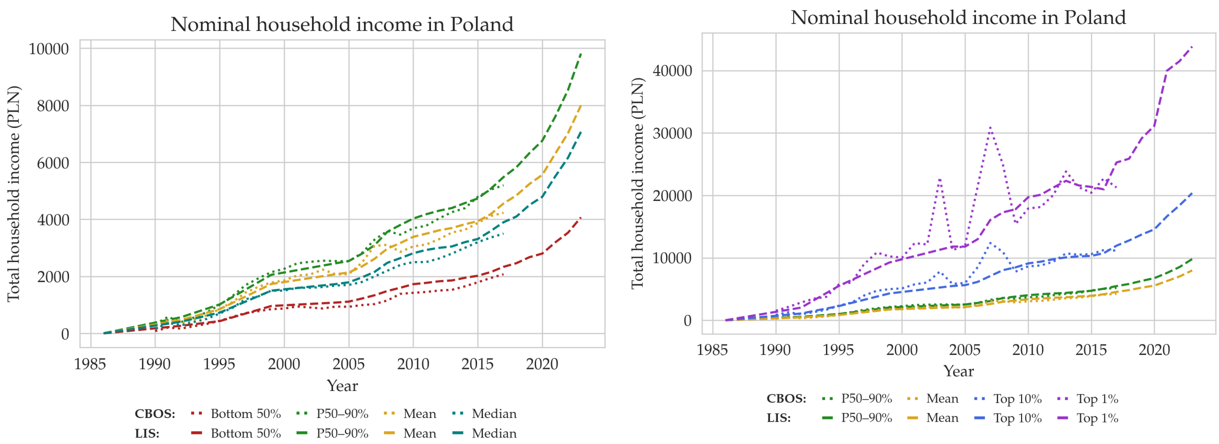
4. Empirical findings

In this section, we first analyse at a national level the evolution of the income groups mean income, as well as their income share, to then dive into geographical decompositions to better understand the income inequality widening over the period.

4.1. Country-level income inequality among income groups evolution

The most striking fact when looking at the mean income evolution by income groups over the last decades is that it has risen for every income groups. This is particularly exaggerated on figure 1 because the nominal income doesn't take into account the inflation that has stricken Poland at the beginning of the 1990's (see appendix 15), as the country opened its economy to the Western investors and drastically reformed its economic and political system. When we look at the real mean income of the income groups, on figure 2, we witness that the real income per income groups haven't gone up as much as the nominal value would suggest but that it has undoubtedly risen. More specifically, we observe that the time at which the the incomes have started to rise, and the gap between income groups to widen (especially with the take-off of the top 1% income group mean income), is soon after the entry of Poland into the European Union (EU) in 2004. This has been also acknowledged by Bukowski and Novokmet (2021), linking it to the large business income component of the top income groups incomes. This can be explained mainly by the fact that the entry into the EU has opened new markets to the Polish firms, and thus has boosted the business incomes of the top income groups.

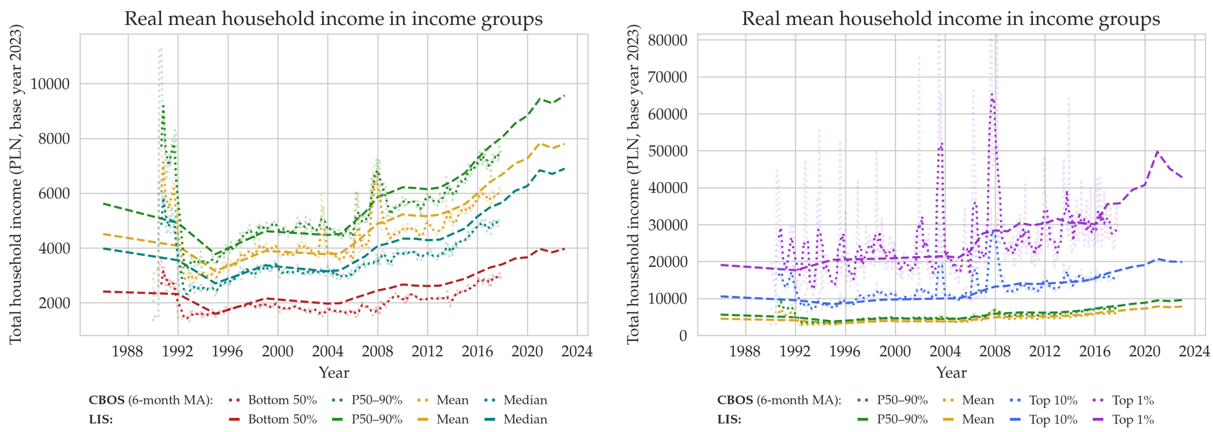
What the figure 2 also allows us to witness is that the top income groups have seen their average income (more) than doubled, when the one of the other income groups have increased, but to a lesser extent. In 1986, according to the LIS data, the real mean monthly income for the middle 40% income group was slightly under 6 000 PLN (in 2023 value), weather in 2023, it was slightly under 10 000 PLN. Similarly, the mean income in the bottom 50% income group has went up from around 2 300 PLN in 1986 to around 4 000 PNL in 2023 ; for the top 10% income group it has went up from 10 000 PLN to 20 000 PLN ; and for the top 1% income group, it has gone up from 20 000 PLN in average to 43 000 PLN. The CBOS data validate these findings, showing however monthly variations at times of economic shocks (hyperinflation in 1991, Subprimes crisis in 2008). Thus, the top income groups have captured a higher portion of the total income growth than the less doted income groups, which the analyse of the income shares confirms.



(a) Among the bottom and the middle income groups

(b) Among the top income groups

Figure 1: Nominal mean household income per month among income groups in Poland (1986-2023)



(a) Among the bottom and the middle income groups

(b) Among the top income groups

Figure 2: Real mean household income per month among income groups in Poland (1986-2023)

These income shares are presented in figure 3. We can see that the income share of the top 1% in our results isn't as high as what Bukowki and Novokmet find (see appendix 16). What is the most probable is that our share of the top 1% income group is underestimated as the LIS and CBOS surveys unintentionally under-sample very top income individuals, less likely to declare their incomes than less wealthy people. Thus the real top 1% income share is certainly rather located between 0.1 and 0.15 between 1990 and 2015. However, we can nevertheless witness that Poland has a strong middle class, even if its share may be slightly overestimated. The CBOS data there allow us to see the impact of some economic shocks on the Polish income shares : the October 2003 austerity plan, including significant cuts in social spending in the perspective to reduce the budget deficit in preparation for accession and Poland's contribution to the EU budget, and the 2008 crisis[5]. As in most of the countries hit by the Subprime crisis, these are the bottom and, to a lesser extend, the middle income group that were the most affected by the crisis. However, it didn't seem to have lead to a permanent change in the trends of the income shares.

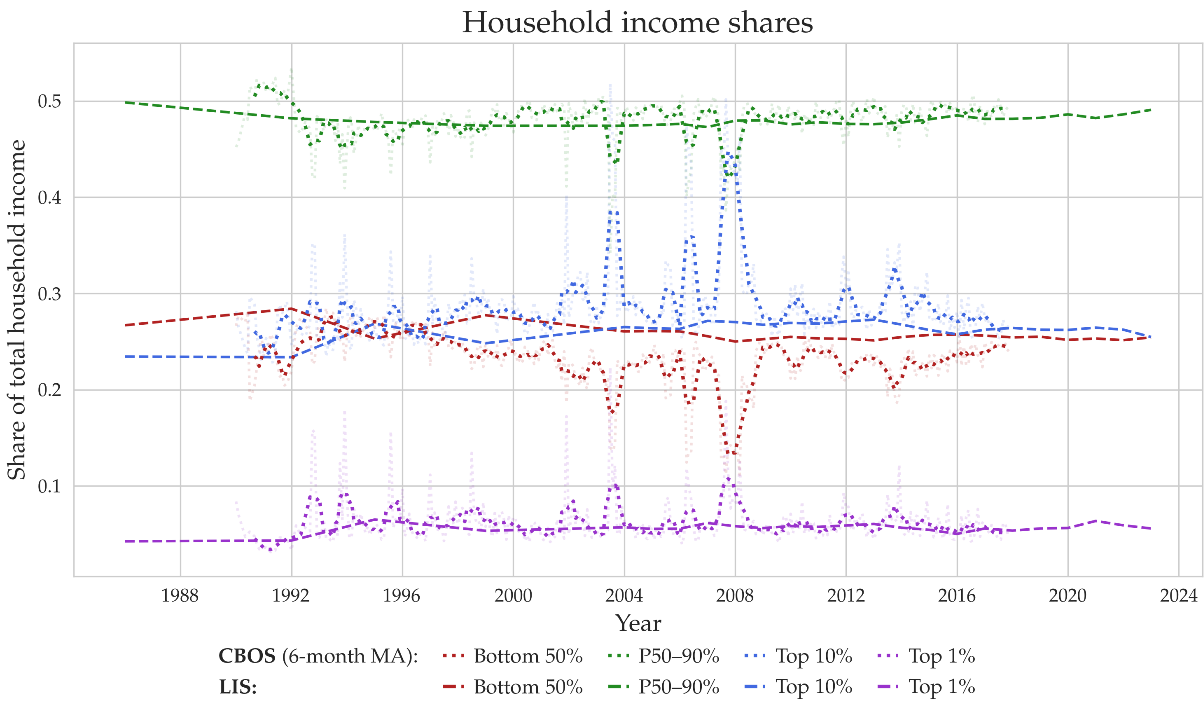


Figure 3: Household income shares in Poland (1986-2023)

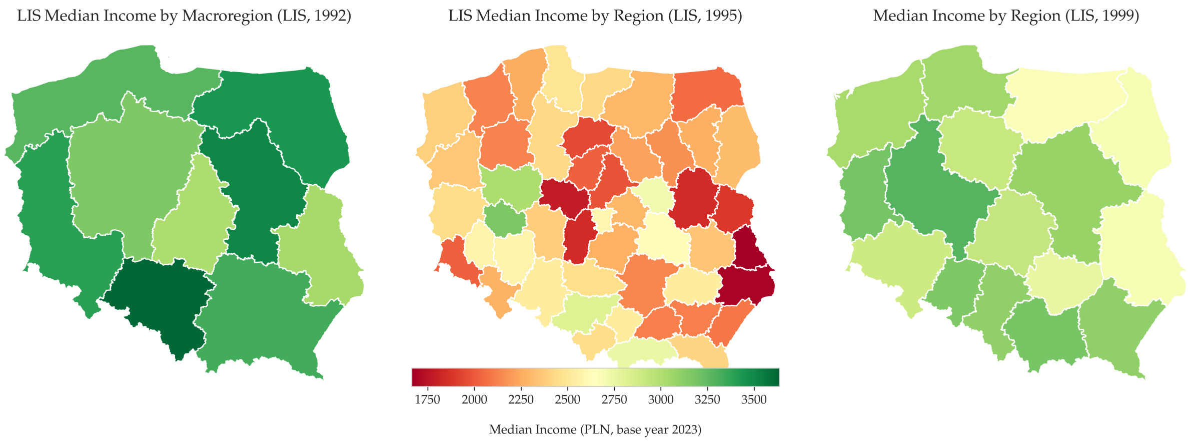
As presented in the introduction, we have insights that these evolutions of income shares and mean income in income groups in Poland is certainly linked to regional disparities. That is what we will uncover in the following section.

[5]Changes in the weighting procedure of CBOS in 2008 may also drive an effect, but it was a too complicated issue to fix for us in the time allowed for the redaction of this paper.

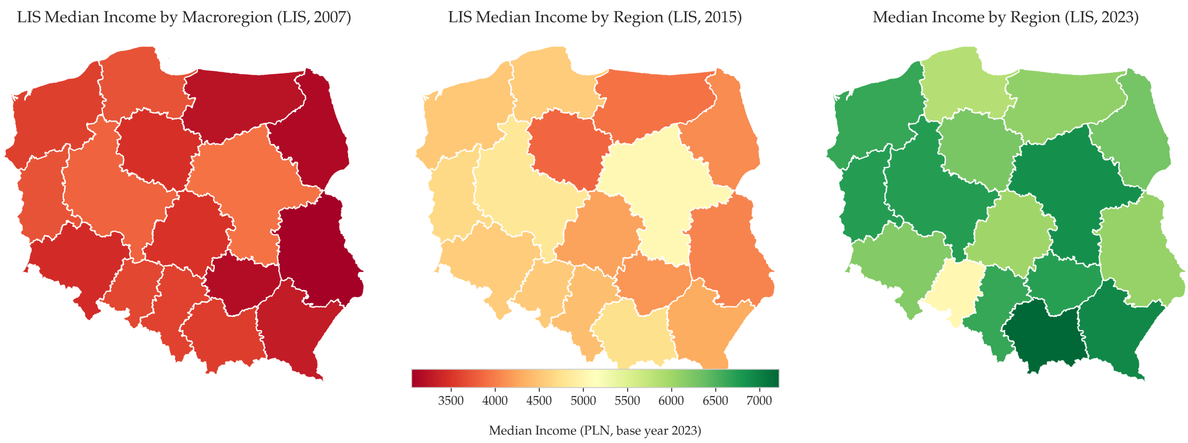
4.2. Regional dynamics explaining the increase in income inequality among income groups

4.2.1. Theil index at the voivoidship level

In traditional public discourse, it is usually assumed that Poland is divided into two seemingly homogeneous parts: poor east and rich west. Nearly always the capital city of Warsaw constitutes a strong outlier on the right part of map. This stereotype can be only partially confirmed if we consider the Figure 4. Due to high spatial aggregation of the early waves of the HBS, we cannot see the exact regional differences in the year 1992. Nevertheless, the macroregions with the lowest median income are located in the middle and eastern parts of the country. Warsaw agglomeration and Silesian conurbation inflate the average income of the two highest-earning macroregions. If we compare this plot with the maps from 1995 or 1999, we can observe clearly the accumulation of low-income regions mostly in the eastern, but also in the central part of the country. The analysis of real income evolution in Poland in early 1990s is in general a difficult task. After the period of hyperinflation, the real



(a)



(b)

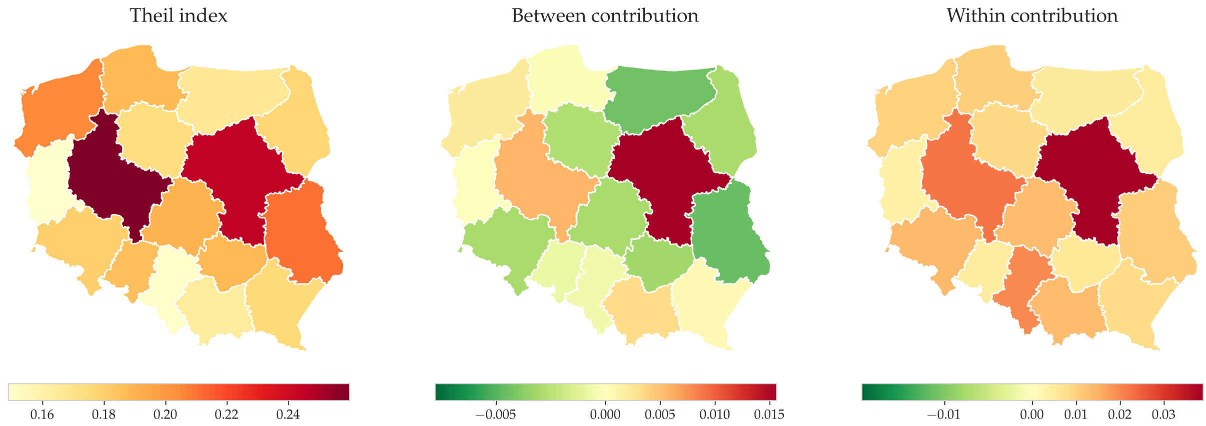
Figure 4: Real median household income per month among across Voivoidships in Poland

median income was high and decreased later in the decade. Yearly CBOS data show a similar dynamics, however it is shifted by a year, suggesting that part of the data from 1992 wave of HBS come from 1991 (see Fig. 13). The median real income rose slightly between 1999 and 2007 in each region. The period of the highest increase of the median can be observed between 2007 and 2023, when the income nearly doubled in each voivodeship. As expected the respondents in eastern voivodeship report on average lower incomes than the western regions.

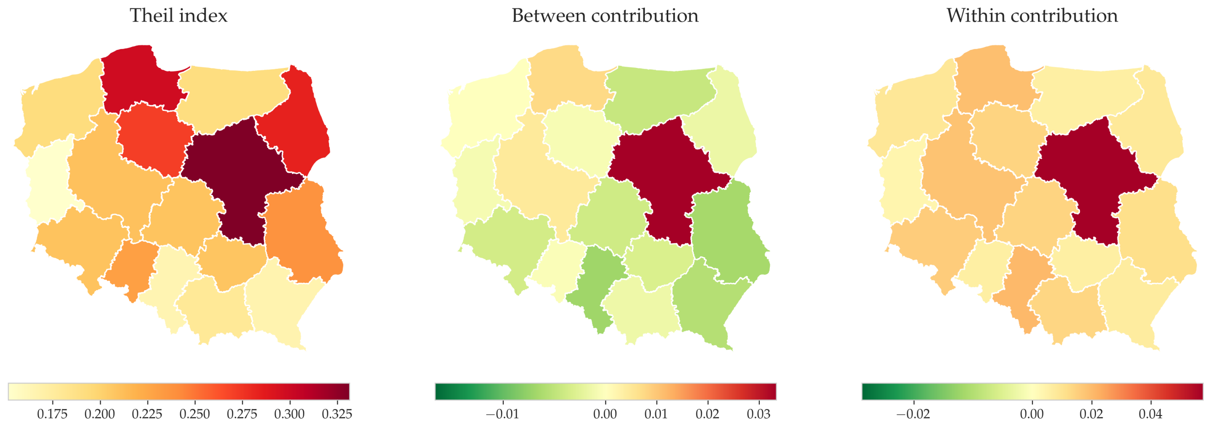
In general, CBOS data shows a similar dynamics and provides stronger evidence for the persistence of the east-west division during the years. On the other hand, CBOS dataset is much more exposed to presence of outliers that have not been removed during the data cleaning procedure and therefore are not presented in this section. Selected CBOS maps are available in the appendix C.2 (Fig. 13).

The one thing that remains unchanged across the years is the fact that the income values for some of the regions are inflated by cities that are a part of them. To investigate this issue closer, we employ the Theil index and compute both the total value and the contributions of each region to its between and within component. In order to conduct a dynamic analysis, we have chosen a set of four different years. 1999 is the first year after the administrative reform, so the data can be grouped by (new) voivodeships and the results get more comparable across years. We choose also 2007, as Bukowski and Novokmet (2024) observe a topping of the top 1% and top 10% income share on that year. We include 2023 as it is the last available wave of the HBS. To establish regular year points, we analyse also the year 2015 as the middle point between 2007 and 2023. We so have 4 year focuses with 8 years between each.

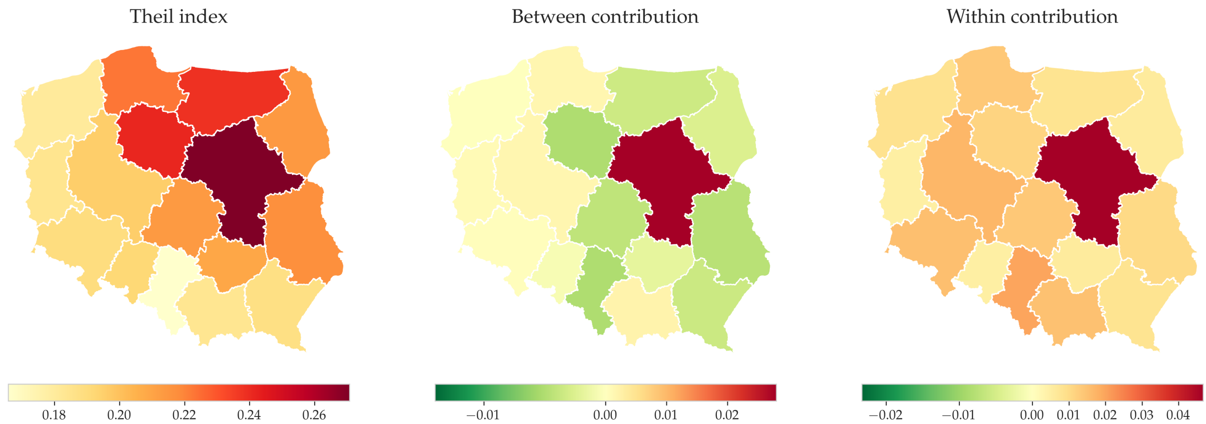
The figures of the panel 5 present a consistent pattern. The Masovian Voivodeship exhibits the highest values of the Theil index in nearly every year. The most notable fact is that this region is both contributing the most to the between and to the within component of the Theil index. The average income of this voivodeship exceeds the values for other regions what leads to high between contribution. According to the values of the within component, the region is very heterogeneous. This finding is expected, since the Polish capital is located in this voivodeship. The high-income city is driving the inequality in the region. Besides Warsaw, this region consist mostly out of rural areas that are exposed to flight of the human capital to the aforementioned city. Other interesting region is the Silesian Voivodeship in southern Poland. It exhibits slightly negative values of the between contribution and high values of the within component. This region is a highly urbanized, post-industrial area which is known for being heterogeneous with respect to standard of living. Eastern regions and regions around the Masovian voivodeship tend to evince both negative values of the between contribution and the low values of the within index.



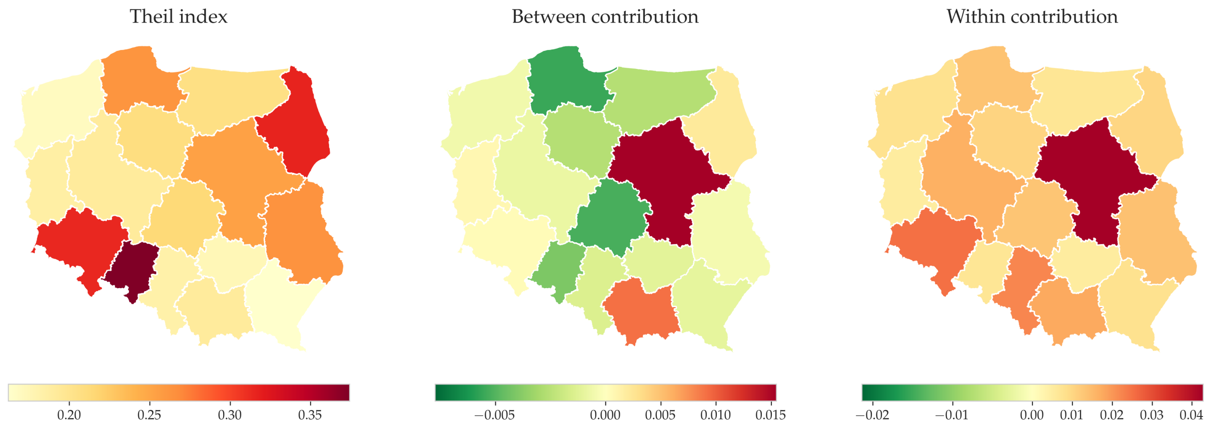
(a) 1999



(b) 2007



(c) 2015



(d) 2023

Figure 5: Theil index and the contributions of voivodeships

This finding suggest that the income of households is on average low and the interiors of these regions are homogenous with respect to that aspect. The final important finding is that the western regions contribute less to the between component with every year. The average incomes in these voivodeships are not growing fast enough to exceed the national average inflated by the Masovian Voivodship – and in fact by Warsaw.

Big cities and highly urbanized regions have proved to bias the analysis on the voivodship level. To further motivate our deeper analysis of the regional income inequalities, we analyse first the findings from the Figure 14d. In this panel we can see the values of the Theil index computed with respect to subdivision into locality classes. As before, we can observe their contributions to its between and within components. Both datasets present a similar dynamic: rural areas have negative between contributions (income lower than national average) and the bigger the city the higher the between contribution. The values of the indices computed from CBOS data have larger volatility, than the one based on LIS data. It can be surprising that rural areas exhibit highest within inequality. In the next section,

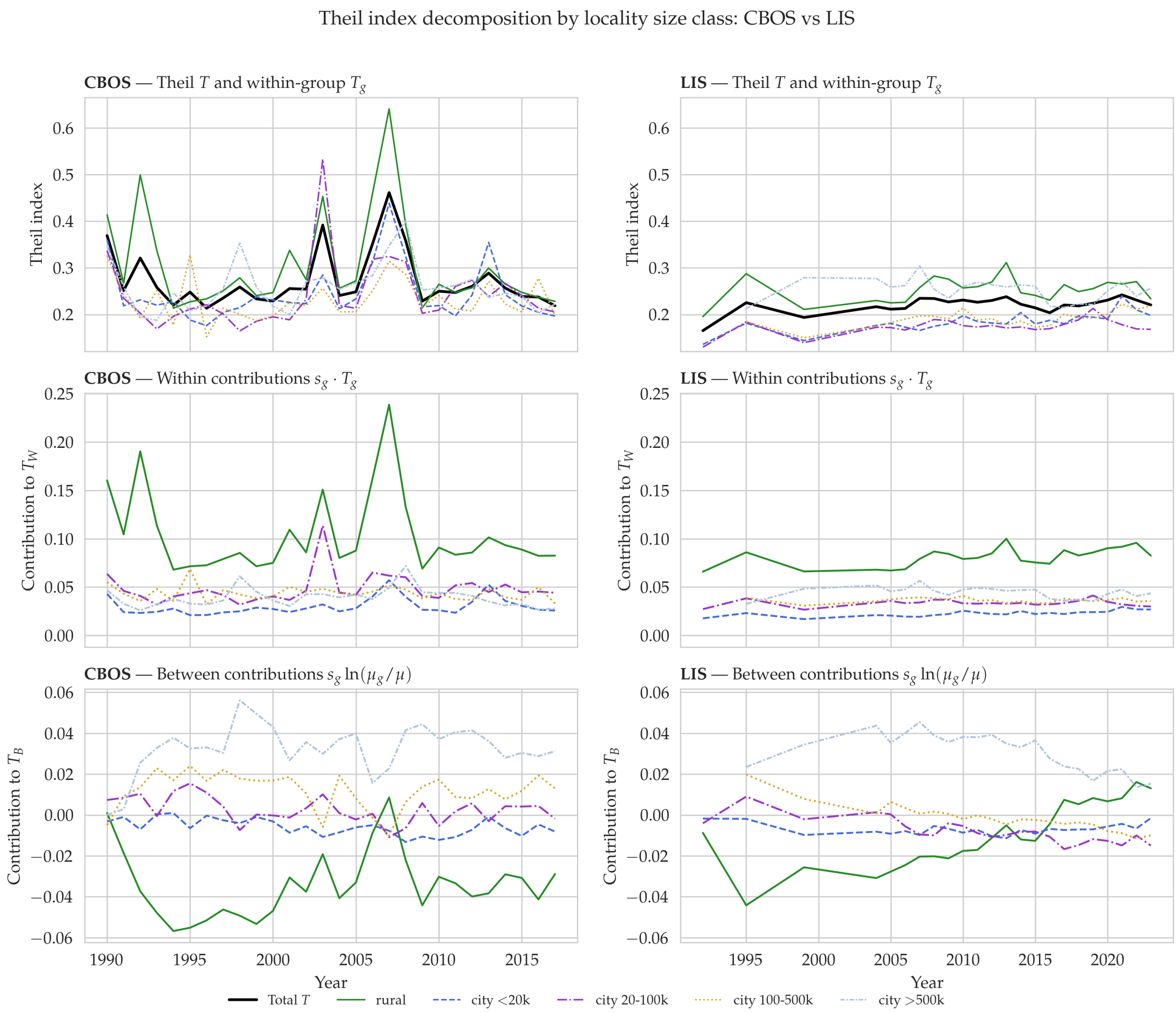


Figure 6: Theil index of household income inequality with respect to locality classes

we will show that we can partially explain it by the voivodship-specific differences between rural areas. An interesting phenomenon can be observed on the bottom right plot, where the rural areas are contributing more to the between index. This finding is consistent with Bukowski and Novokmet (2024), who have proved that the rich individuals tend to now live in rural areas more than in the past. We will investigate this issue further in the next section.

4.2.2. Theil index of the locality groups

In this section, we will focus on the analysis of the location groups, i.e. locality classes inside different voivodships. The shapes visible on the map are in the majority cities between 100.000 and 500.000 inhabitants and greater. Due to data availability issues, it is not possible to analyse the rural areas separately in a rigorous way[6].

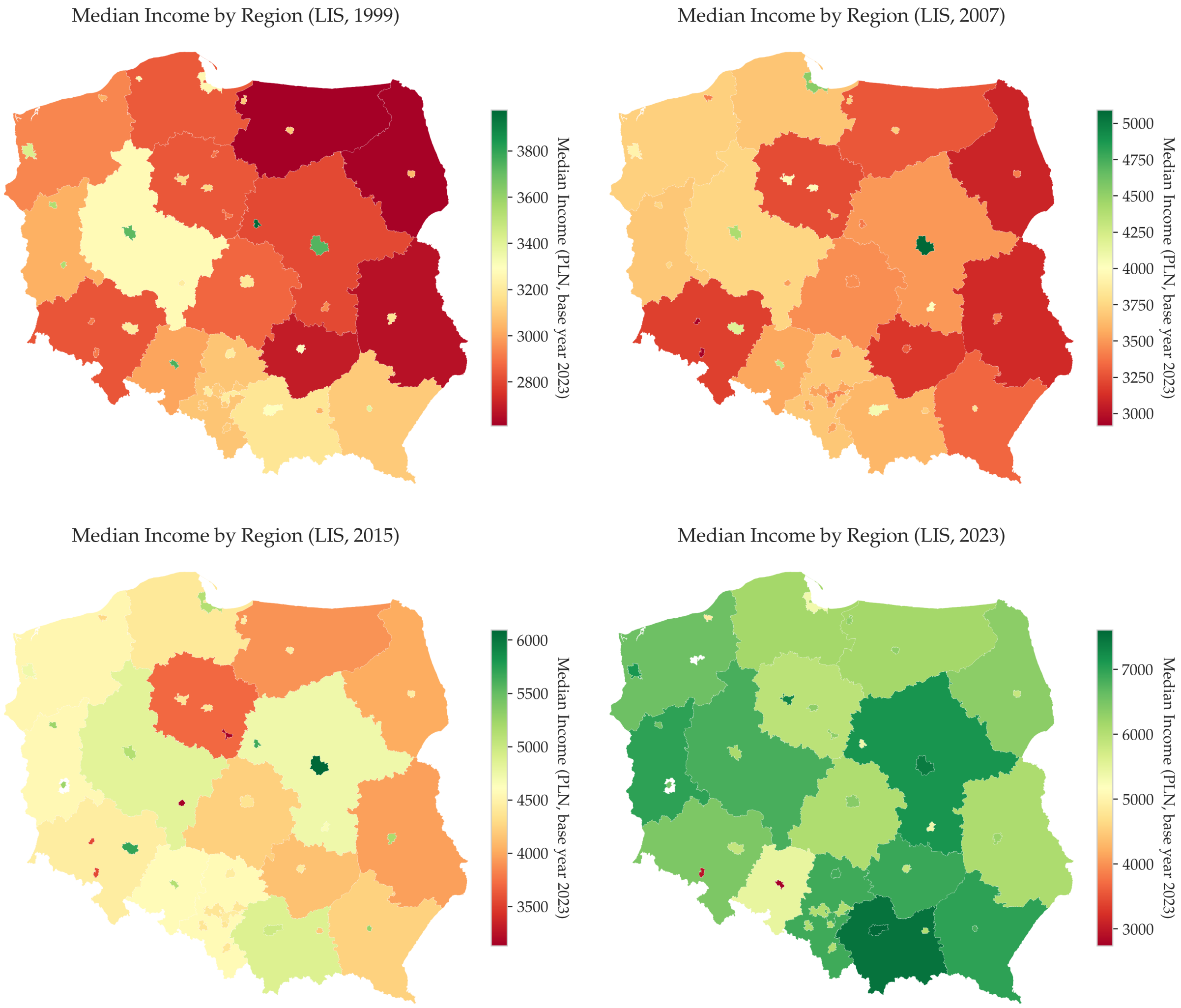


Figure 7: Real median household income per month among across Voivoidships in Poland (selected years)

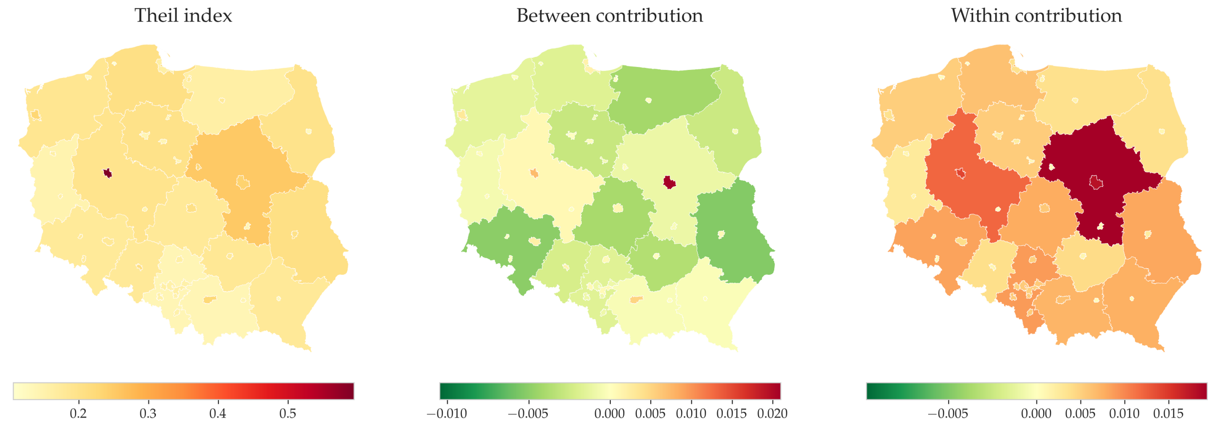
[6]To conduct such an analysis one would have to use census microdata to distinguish rural areas that are part of larger agglomerations from the *true* rural areas and build cross-tabulations for these locations

Before we investigate the Theil index, we will briefly describe the evolution of the median. As expected, large cities exhibit significantly larger values of the median household income, than the rest of the voivodships. There is a group of cities that seem to have median income of similar or even smaller magnitude as their regions. This cities are located in post-industrial regions of Upper and Lower Silesia. Most of these cities declined either during or after the transformation period when unprofitable workplaces were closed down[7] Finally, one observe an interesting dynamic: the differences between the median in big cities and their regions is becoming smaller with time. This raises the question whether Poland is becoming more spatially equal with respect to household income?

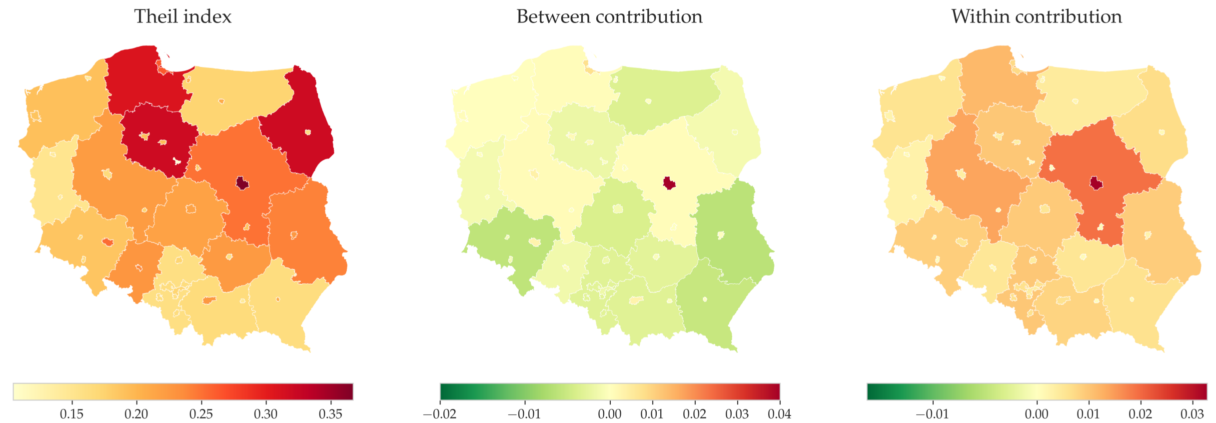
To answer the naive question from the previous section, we employ the Theil index. For the first three timestamps of our analysis, we can clearly see that Warsaw is contributing the most to the between component of the Theil index. The capital city is followed by other large towns, however their relative contributions decrease with time. This suggests that income in other cities is rising slower than in Warsaw. For the within contributions, we can observe an interesting phenomenon. All cities other then Warsaw are characterized by a rather low level of within inequality, while their regions tend to be more unequal. The most unequal region is consistently the Masovian Voivodeship. One of possible explanation for that could be the fact that parts of the high-income Warsaw agglomerations are not separated from the low-income Masovian rural areas. This reason could be also an explanation for the high between contribution in 2023 of this region. Suburbanization and urban-sprawl are currently one of the biggest problems of the agglomeration. High-income well-educated Varsovians tend to move out of the city while still working in the capital. Similar results can be observed for the agglomeration of Cracovia and the Silesian conurbation. This dynamics can be closer investigated with the maps in the appendix C.2 (Fig. 14).

When comparing 1999 with 2023 the country has become on average less unequal. The lowest values of the index can be observed around year 2015, what would imply that we can expect an upward sloping trend in that matter. Throughout the presented spatial income inequality analysis, one could observe that the inequality in Poland was inflated mostly by the within components. The results of this section show that this trend can slightly change in the future. Based on our locality groups, the locations are becoming heterogenous when comparing them to each other, but their internal structure becomes more homogenous. Separating the whole agglomerations from the rest of their region could be the subject of further researches to better understand the spatial inequality between locality groups.

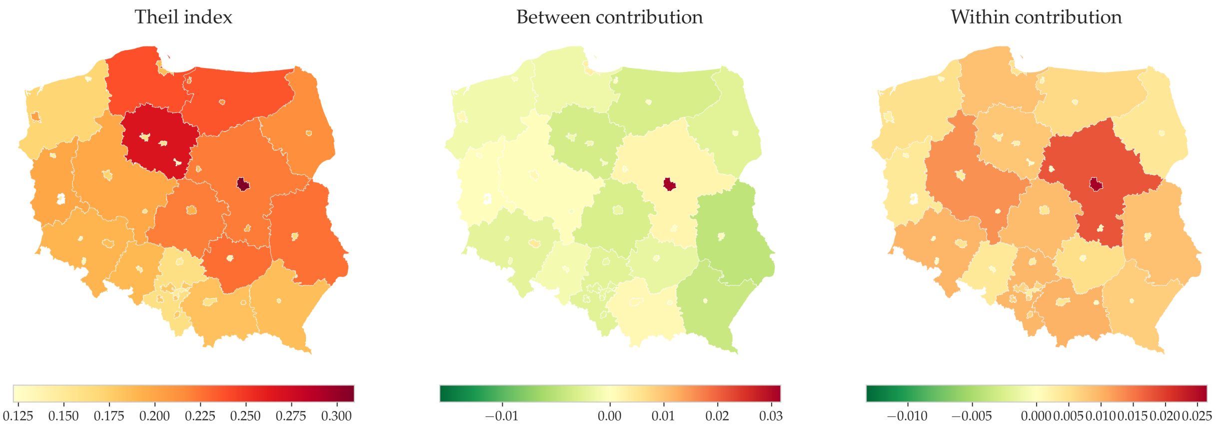
[7] Wałbrzych (Lower Silesia, near Czech border) is the most well-known example of such city.



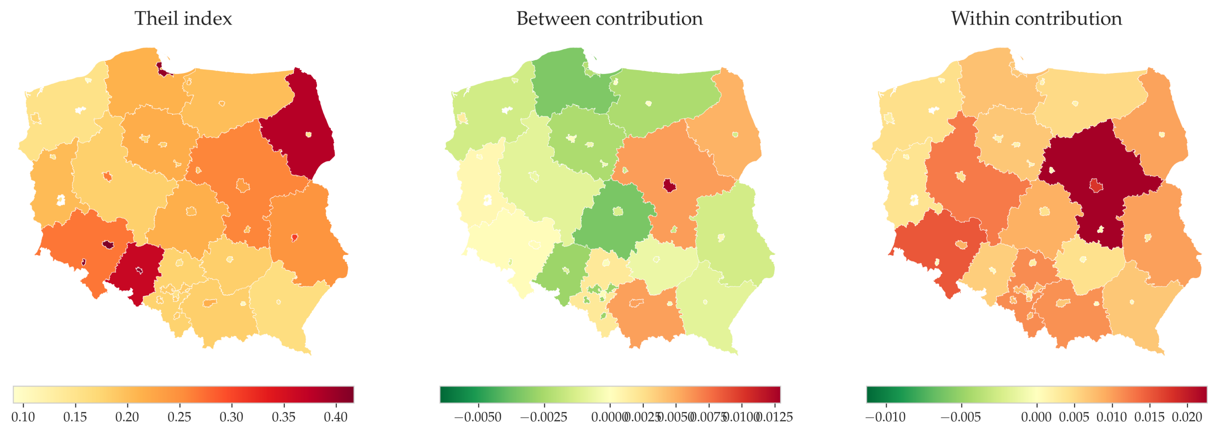
(a) 1999



(b) 2007



(c) 2015



(d) 2023

Figure 8: Theil index and the contributions of the locality groups

5. Conclusion

This paper examines the evolution of spatial income inequalities in Poland from the transition period following the fall of communism in 1990 through 2023. The research aims to determine the extent to which regional disparities and urban-rural divides have driven the significant rise in income inequality, which has seen Poland shift from one of Europe's most egalitarian nations to one of its most unequal.

To conduct this analysis, the study utilizes two datasets, both very carefully reweighted using a regional weighting framework. Albeit we addressed significant missingness in CBOS income variables, this data doesn't prove high interest for the spatial analysis. It nevertheless gives interesting details on the monthly variation of households' income that no other income data sources for Poland can do.

The empirical results demonstrate that while a persistent East-West divide—rooted in the historical legacies of the Prussian partitions—remains visible, this broad regional categorization is insufficient to explain the modern dynamics of Polish income inequality. Instead, the findings reveal that contemporary disparities are primarily driven by the intense concentration of wealth within major urban centers, most notably Warsaw. This phenomenon is best illustrated by the Masovian Voivodeship, which exhibits the nation's highest Theil index values; here, the concentration of the business activity generates profound "within-region" disparities between the affluent metropolitan core and its rural periphery. On that point, it seems important to me to recall one last time the analysis of Bukowski et al. (2023), stressing the role of the political choices into the shaping of the low taxation on business income. Furthermore, our study identifies Poland's 2004 EU accession as a pivotal structural turning point, after which the growth in top income shares was disproportionately captured by urban elites and business-income groups. Ultimately, the analysis concludes that "within-locality" and "within-region" imbalances have superseded inter-regional differences as the dominant contributors to Poland's national income inequality.

A. Administrative and statistical division of Poland

Level	Sublevel	NUTS	Original name	English name	Administrative?
0		0	Polska	Poland	✓
1		1	Makroregion	Macroregion	
2			Województwo	Voivodeship	✓
3		2	Regiony	Regions	
4		3	Podregiony	Subregions	
5			Powiat	County	✓
6			Gmina	Commune	✓
6	1		Gmina miejska	Urban commune	✓
	1.8		Dzielnica/delegatura	District of a city	✓
	1.9		Dzielnica m.st. Warszawy	District of Warsaw	✓
6	2		Gmina wiejska	Rural commune	✓
6	3		Gmina miejsko-wiejska	Urban-rural commune	✓
	3.4		Obszar miejski	Urban area	
	3.5		Obszar wiejski	Rural area	
7			Miejscowość	Settlement	✓

Table 1: Administrative and statistical division of Poland after the administrative reform of 1999

B. Statistical Appendix

B.1. Descriptive statistics

Table 2: Descriptive statistics — LIS Poland person files, all available years

Panel A: Sample size, sex composition, and age distribution

		Sex (%)		Age (years)		
Year	N	Male	Female	P25	Median	P75
1986	34,201	47.1	52.9	14	33	52
1992	18,807	46.4	53.6	15	36	55
1995	103,530	47.8	52.2	15	34	49
1999	99,791	47.7	52.3	16	34	50
2004	99,038	47.8	52.2	18	35	52
2005	107,124	47.7	52.3	18	35	52
2006	114,311	47.9	52.1	18	36	53
2007	111,992	47.8	52.2	18	36	53
2008	109,819	47.7	52.3	19	37	54
2009	108,038	47.8	52.2	19	38	55
2010	107,967	47.9	52.1	20	38	56
2011	107,239	47.9	52.1	20	38	56
2012	105,327	47.7	52.3	20	39	57
2013	102,780	47.9	52.1	21	39	57
2014	101,669	47.6	52.4	21	40	58
2015	101,076	47.6	52.4	21	40	59
2016	99,230	47.7	52.3	22	41	60
2017	97,434	47.7	52.3	22	42	60
2018	95,472	47.8	52.2	22	42	61
2019	93,674	47.7	52.3	22	43	62
2020	87,603	47.8	52.2	22	43	62
2021	78,726	47.6	52.4	24	44	62
2022	70,126	47.2	52.8	24	45	63
2023	68,329	47.4	52.6	24	45	63

Notes: Sex proportions are conditional on non-missing sex.

Panel B: Education distribution (% of observations with non-missing education)

Year	Primary & below	Vocational secondary	General secondary	Post-secondary & tech.-voc.	Tertiary	Missing (%)
1986†	9.5	24.7	60.5	0.0	5.2	34.5
1992	36.8	28.3	25.2	3.9	5.8	24.4
1995	30.2	30.9	27.8	5.0	6.1	24.1
1999	34.9	29.0	7.9	21.1	7.0	22.3
2004	32.3	27.1	8.6	21.9	10.2	15.6
2005	32.1	26.8	8.7	22.1	10.2	15.8
2006	31.5	26.9	8.9	22.1	10.7	15.4
2007	30.4	26.7	9.4	22.1	11.4	15.2
2008	29.6	26.5	9.6	21.9	12.3	14.8
2009	28.5	26.5	9.8	22.0	13.3	14.6
2010	27.2	27.1	10.0	20.8	14.9	14.6
2011	26.3	27.1	9.7	21.1	15.9	14.6
2012	25.6	27.3	9.5	21.2	16.4	14.7
2013	24.6	27.1	9.7	21.1	17.7	15.1
2014	23.9	26.9	10.0	21.0	18.1	14.8
2015	23.3	27.0	9.6	21.6	18.5	14.9
2016	22.4	27.4	9.6	21.8	18.9	14.8
2017	21.5	27.4	9.6	22.0	19.5	15.0
2018	20.7	27.2	9.7	22.6	19.8	15.0
2019	20.1	26.8	9.8	22.3	21.0	15.2
2020	17.7	25.6	9.8	23.5	23.4	16.2
2021	16.4	24.5	9.6	24.8	24.6	15.3
2022	16.3	24.2	9.6	24.9	25.0	15.0
2023	16.1	24.0	9.7	24.5	25.7	14.8

Notes: Education proportions are conditional on non-missing highest educational level. The five categories correspond to: **Primary & below** — *podstawowe i niższe*; **Vocational secondary** — *zasadnicze zawodowe / branżowe*; **General secondary** — *średnie ogólnokształcace*; **Post-secondary & tech.-voc.** — *policealne oraz średnie zawodowe / branżowe*; **Tertiary** — *wyższe.* † In 1986 the source variable does not distinguish general from post-secondary education; all secondary-level observations are assigned to General secondary, so Post-secondary & tech.-voc. equals 0%.

Table 3: Descriptive statistics grouped by survey year - CBOS survey

Panel A: Sample size, sex composition, age distribution, and household size

			Age (years)			Household size		
Year	*N*	Female (%)	P25	Median	P75	P25	Median	P75
1990	16,086	50.1	32	42	56	2	4	4
1991	13,390	52.1	31	42	58	2	4	4
1992	9,193	54.9	34	44	61	2	3	5
1993	14,132	54.8	35	45	62	2	3	4
1994	13,195	55.5	35	45	62	2	3	4
1995	14,153	55.5	35	46	62	2	3	4
1996	13,816	55.7	35	45	62	2	3	4
1997	14,821	56.1	34	45	61	2	3	4
1998	13,887	56.0	35	46	62	2	3	4
1999	13,073	55.9	35	47	62	2	3	4
2000	13,171	55.4	34	47	62	2	3	4
2001	12,467	56.1	33	47	62	2	3	4
2002	12,627	56.1	34	47	62	2	3	4
2003	12,799	55.2	33	47	62	2	3	4
2004	11,823	52.9	31	47	60	2	3	4
2005	12,418	53.5	31	47	59	2	3	4
2006	11,988	53.3	32	47	60	2	3	4
2007	11,455	52.3	32	49	61	2	3	4
2008	13,021	54.0	31	48	61	2	3	4
2009	12,742	54.2	32	49	61	2	3	4
2010	11,940	53.5	32	49	62	2	3	4
2011	12,712	54.0	33	49	62	2	3	4
2012	10,225	54.5	33	49	62	2	3	4
2013	11,522	54.3	33	50	62	2	3	4
2014	11,952	53.9	34	50	63	2	3	4
2015	12,266	54.5	34	51	63	2	3	4
2016	12,392	53.6	35	51	64	2	3	4
2017	12,070	54.8	36	52	65	2	3	4

Notes: Sex proportions are conditional on non-missing sex.

Panel B: Education distribution (% of observations with non-missing education)

Year	Vocational secondary	General secondary	Post-secondary & tech.-voc.	Tertiary
1990	23.3	13.3	21.4	8.1
1991	25.0	12.0	18.3	6.9
1992	24.8	10.5	19.1	7.0
1993	24.8	10.8	18.9	7.1
1994	24.9	10.7	20.7	7.4
1995	24.5	11.0	20.8	7.6
1996	24.8	11.7	22.4	8.4
1997	25.3	12.0	23.1	9.4
1998	25.3	11.9	22.1	9.0
1999	24.5	12.9	22.9	9.4
2000	24.8	13.0	23.5	10.3
2001	26.1	13.3	22.8	10.7
2002	25.2	13.1	23.7	11.6
2003	25.1	13.6	23.3	11.9
2004	26.3	14.4	22.9	11.2
2005	27.5	14.5	22.9	9.3
2006	25.8	15.3	23.2	12.0
2007	25.6	15.2	23.4	13.0
2008	25.0	14.2	22.6	14.9
2009	25.6	12.4	21.4	15.4
2010	25.6	13.0	21.6	17.4
2011	25.1	13.2	22.4	17.0
2012	25.2	12.9	23.1	19.6
2013	24.7	12.5	23.6	19.6
2014	24.5	12.5	23.2	21.8
2015	24.9	11.7	23.7	23.1
2016	23.4	11.6	24.1	25.8
2017	24.5	12.4	24.6	22.9

Notes: Education proportions are conditional on non-missing highest educational level. The omitted baseline category is **Primary & below**.

B.2. Missingness in CBOS data

Test	Target	N	Med. AUC	Med. p	$p < 0.05$	$p < 0.01$	$p < 0.05/N$
LRT Model A (Demogr.)	`income_p`	207	0.672	$\ll 0.0001$	88.9%	82.6%	71.0%
LRT Model B (Demogr. + SOL)	`income_p`	203	0.673	$\ll 0.0001$	93.6%	88.2%	75.9%
LRT Model A (Demogr.)	`income_hh`	313	0.626	< 0.0001	82.7%	74.8%	58.1%
LRT Model B (Demogr. + SOL)	`income_hh`	312	0.636	< 0.0001	84.9%	76.6%	61.5%
Little (1988)	`income_p`	207	—	$\ll 0.0001$	92.3%	85.5%	74.4%
Little (1988)	`income_hh`	327	—	< 0.0001	82.9%	77.1%	61.2%

Notes: LRT = per-survey likelihood-ratio test from logistic regression; Little's = per-survey global χ^2 test.

Table 4: MCAR diagnostic tests for personal and household income.

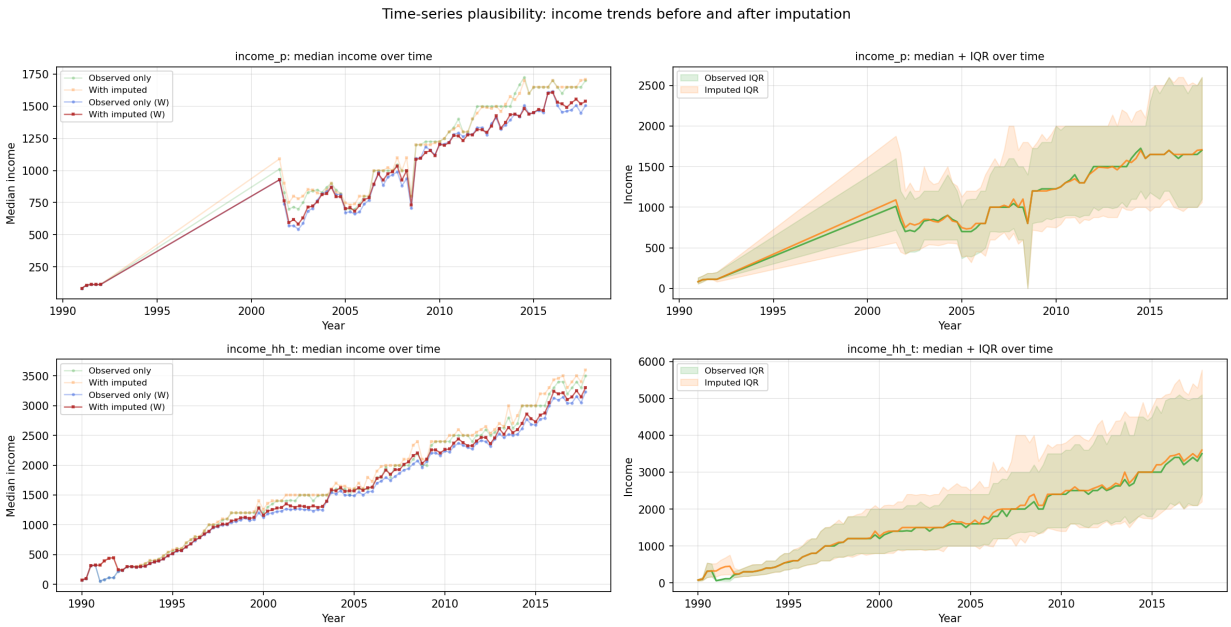


Figure 9: The evolution of mean and the confidence intervalls after imputation.

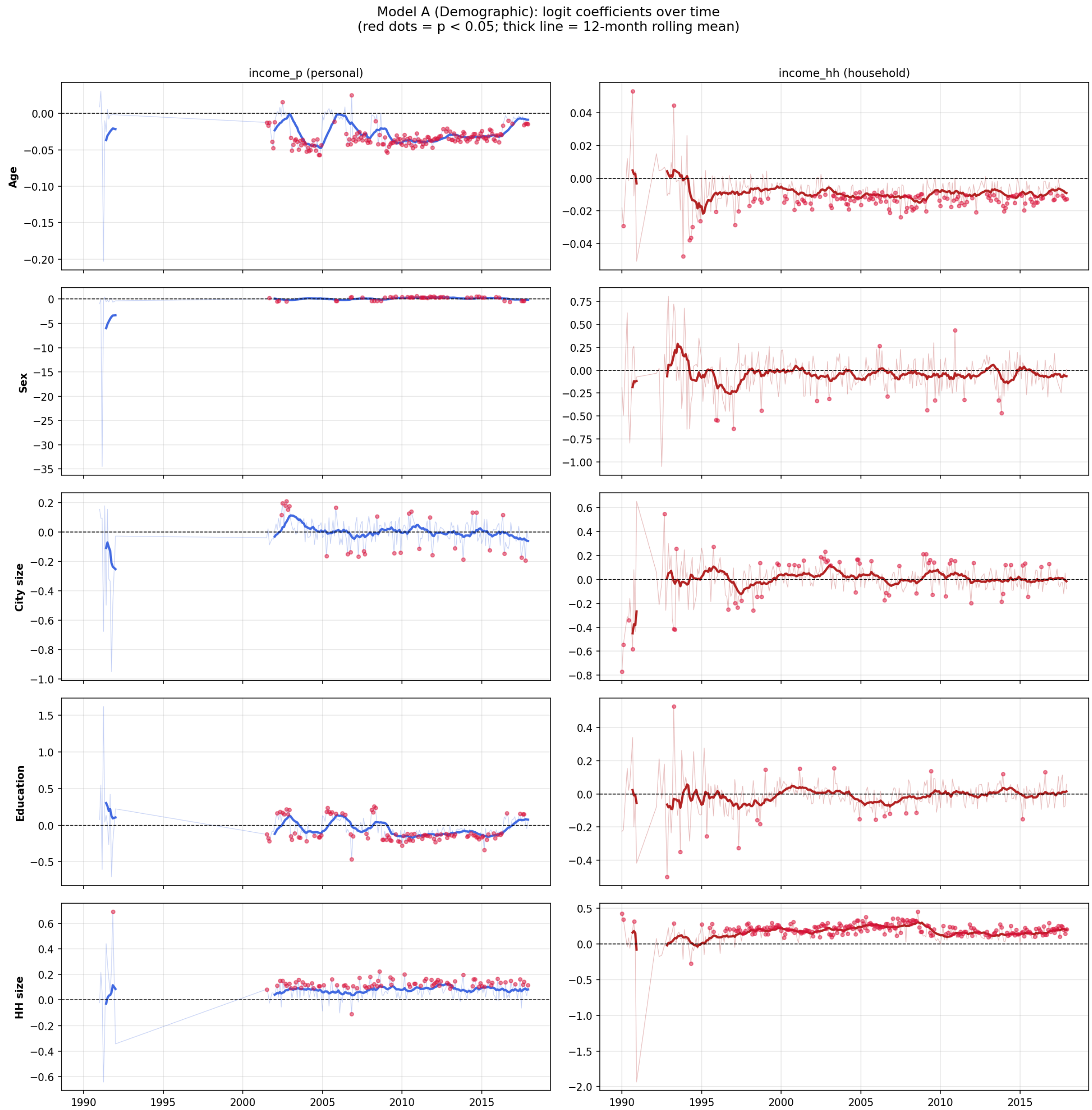


Figure 10: Coefficients of the missingess prediction via logit model.

B.3. Additional demographic data

Data source	Tabulations	Time range	Territorial level	Notes
National Census 1988		1988	6	
	Age *vs* Gender	1988	6	Age, threshold every 10 years
	Gender *vs* Education	1988	6	Education, 4 types
	Household size	1988	6	Size, 4 thresholds
National Census 2002		2002	6	
	Age *vs* Gender	2002	6	Age, threshold every 5 years
	Gender *vs* Education	2002	6	Education, 7 types
	Household size	2002	6	Size, 5 thresholds
National Census 2011		2011	6	
	Age *vs* Gender	2011	6	Age, threshold every 5 years
	Gender *vs* Education	2011	6	Education, 9 types
	Household size	2011	6	Size, 5 thresholds
National Census 2021		2021	6	
	Age *vs* Gender	2021	6	Age, threshold every 5 years
	Gender *vs* Education	2021	6	Education, 9 types
	Household size	2021	6	Size, 5 thresholds
LDB	Age *vs* Gender	1995-2024	6	Age, threshold every 5 years
BAEL	Gender emphvs Education	1995-2024	2 and 3	Education, Education, 7 types

Table 5: Overview of the demographic data for regional weighting.

C. Graphical appendix

C.1. Regional characteristics

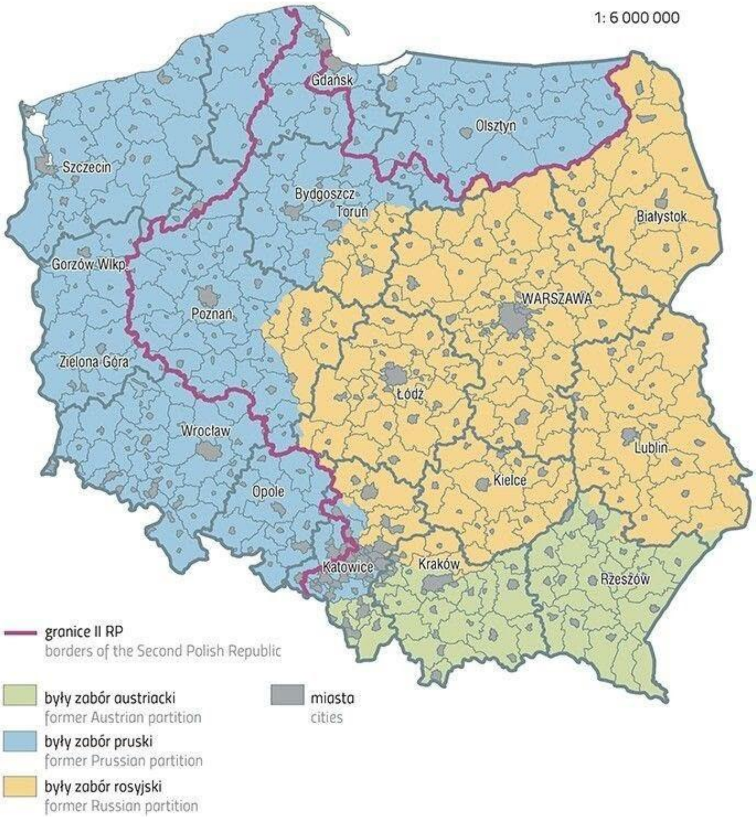


Figure 11: Partition of Poland from 1866 to 1914

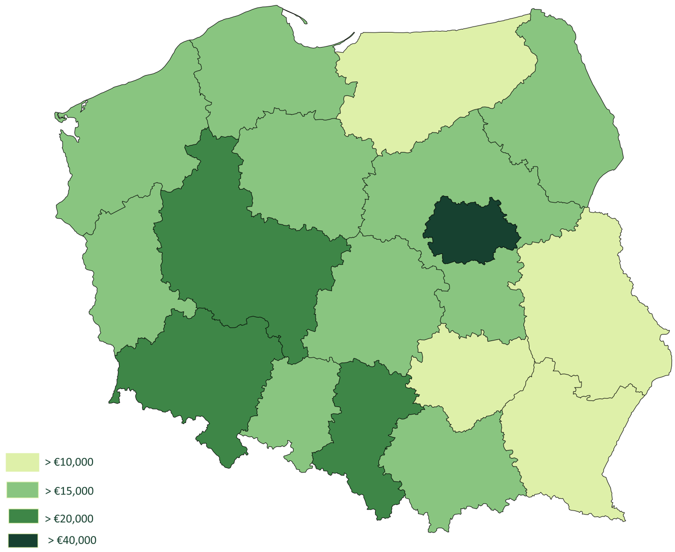


Figure 12: Partition of Poland from 1866 to 1914

C.2. Additional plots and maps

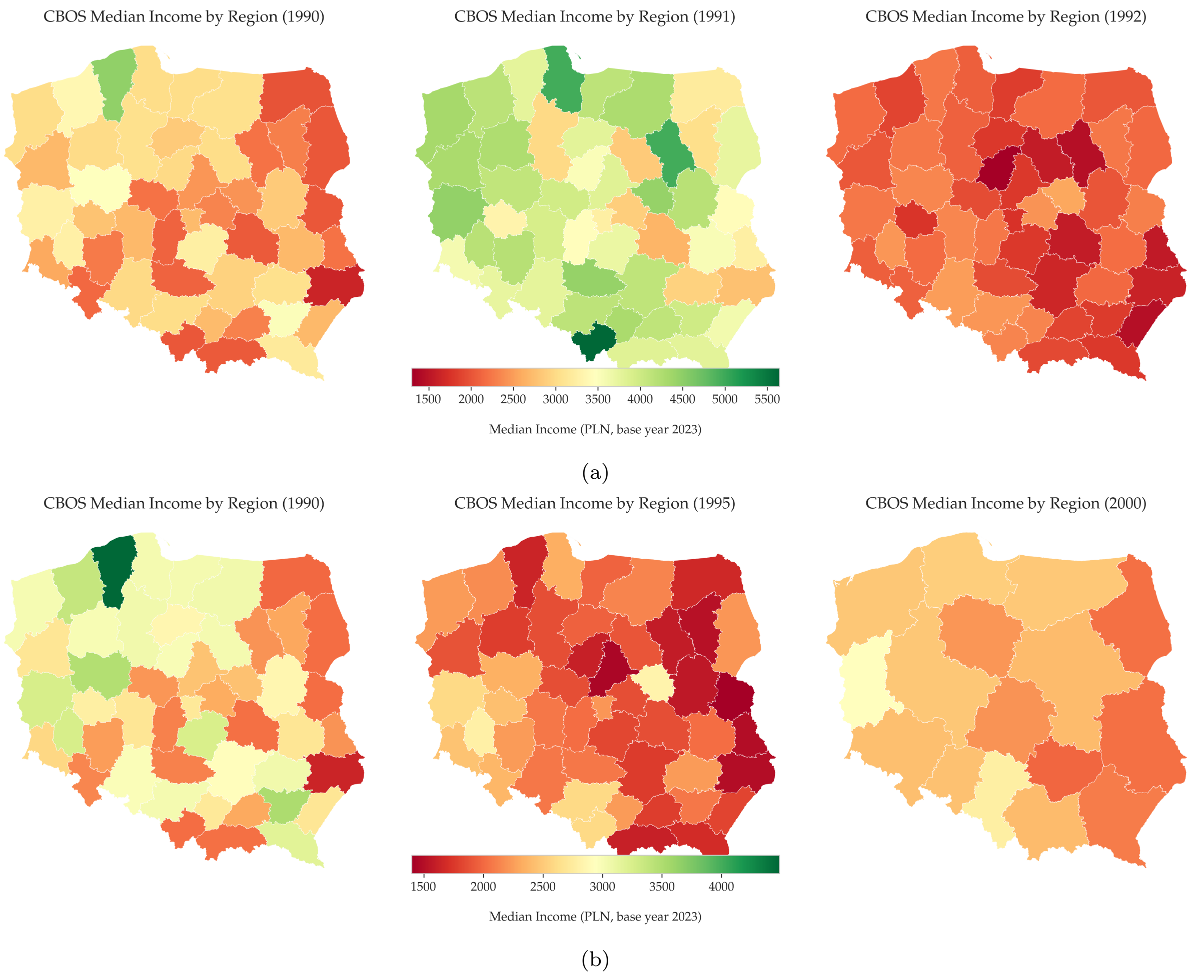
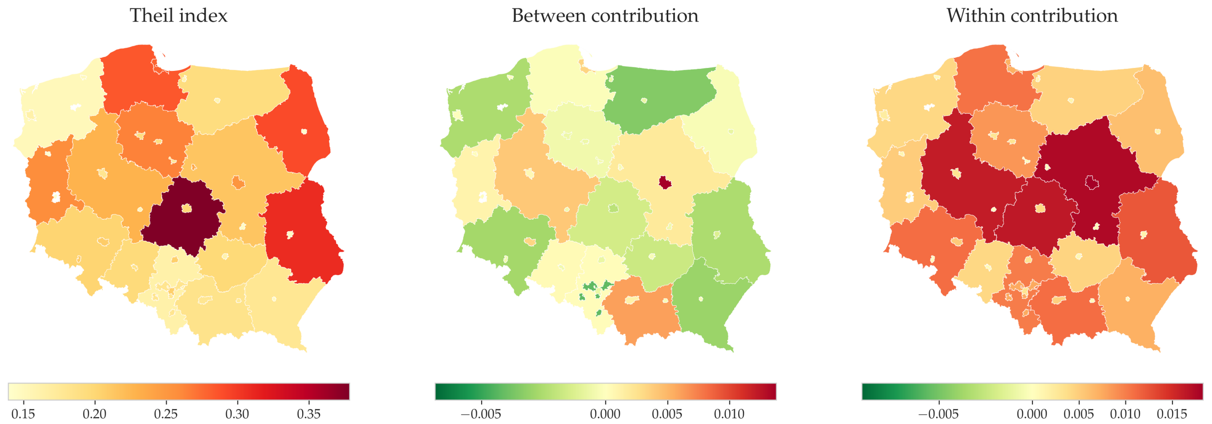
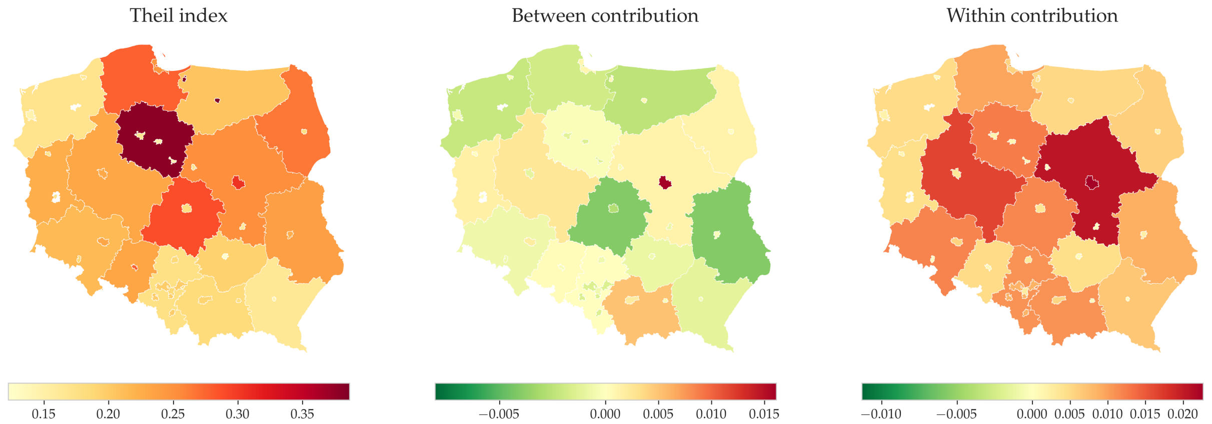


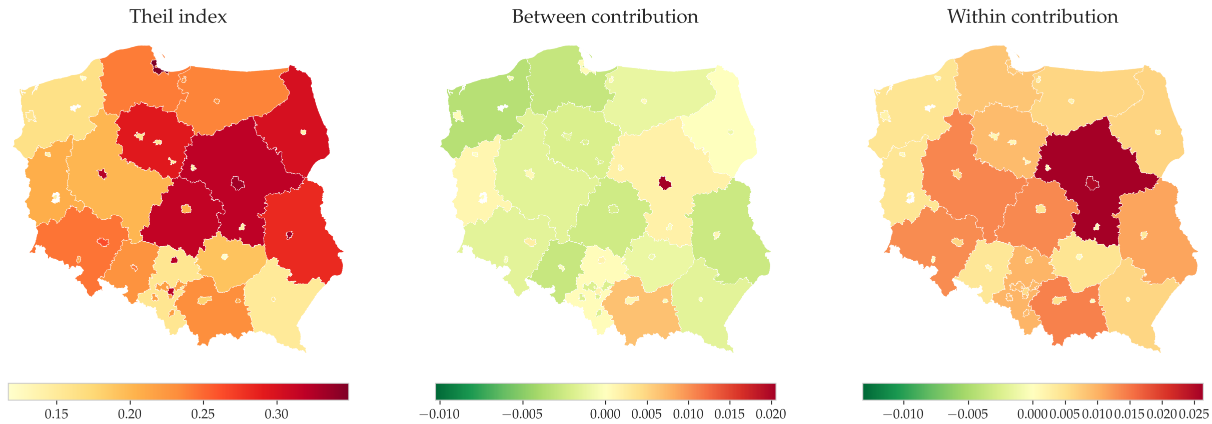
Figure 13: Real median household income per month among across Voivoidships in Poland (CBOS, selected)



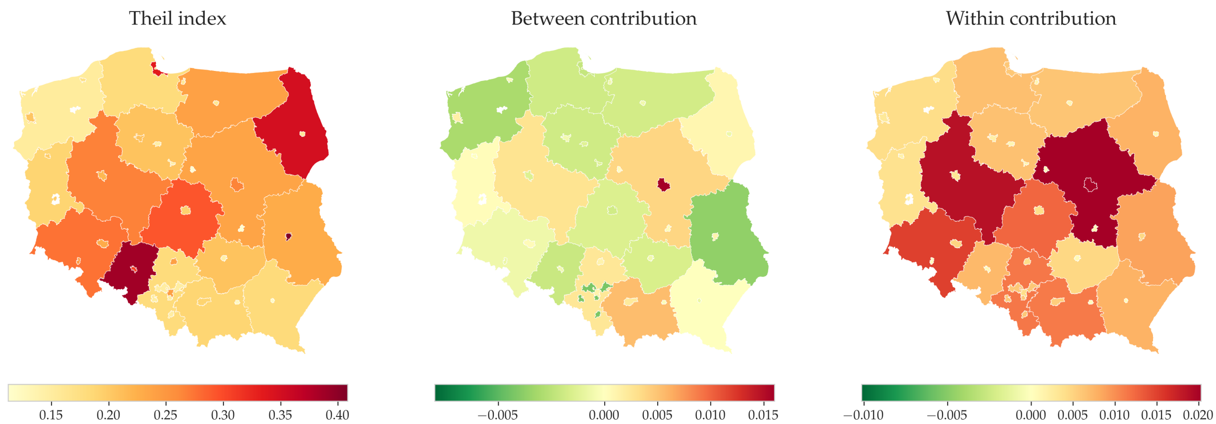
(a) 2019



(b) 2020



(c) 2021



(d) 2022

Figure 14: Theil index and the contributions of the locality groups (LIS, selected)

C.3. Auxiliary plots

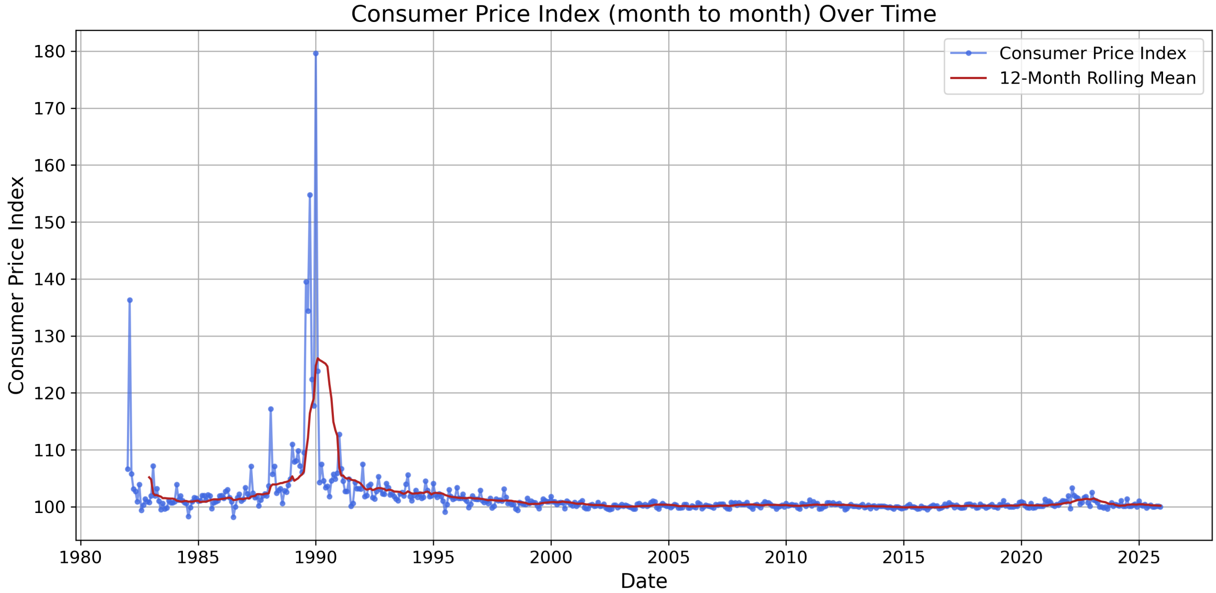


Figure 15: Consumer Price index (1980's-2026)

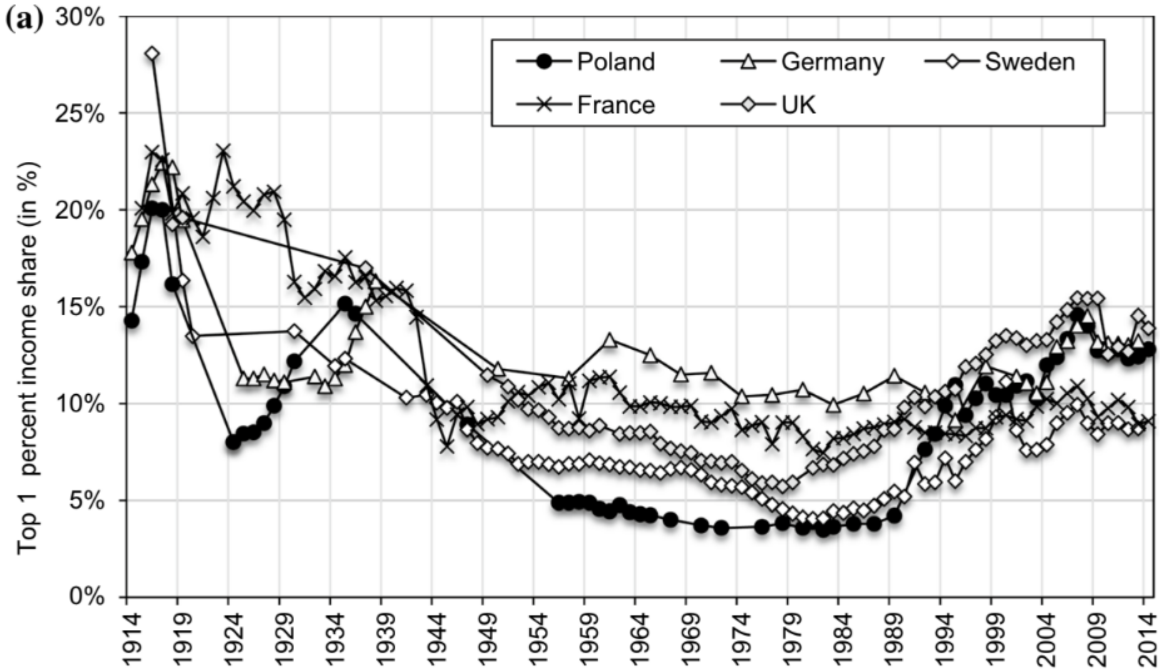


Figure 16: Top 1% income share in Poland, Germany, France, Sweden and the UK, 1914–2014 (source : Bukowski and Novokmet (2021)

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Declaration

We hereby certify that the above term paper has been prepared by me independently without unauthorized external assistance and without the use of any aids other than those indicated, and that we have marked all passages taken verbatim or in substance from published or unpublished writings as such.

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